

Employee Details Analysis

Project Overview

The project uses the "Employee Details" dataset, a company HR dataset containing 150 employee records across 9 original columns.

- Employee ID - unique identifier for each employee
- Name - employee full name
- Department - Finance, HR, Sales, IT, Marketing
- Country - employee work location (Germany, USA, India, UK, Australia)
- Salary - annual salary (numeric)
- Date of Joining - date the employee joined the company
- Performance Rating - rating scale 1 to 5
- Gender - Male, Female, Other, Unknown
- Email - employee contact email

This dataset was chosen because it combines numeric fields (Salary, Rating), date fields (Date of Joining) and categorical fields (Department, Country, Gender), which allows the full range of Power BI features - aggregation, time intelligence, ranking, hierarchies and interactive visuals - to be demonstrated.

Data Cleaning Steps (Power Query)

The raw dataset had several data quality issues that were resolved before modelling:

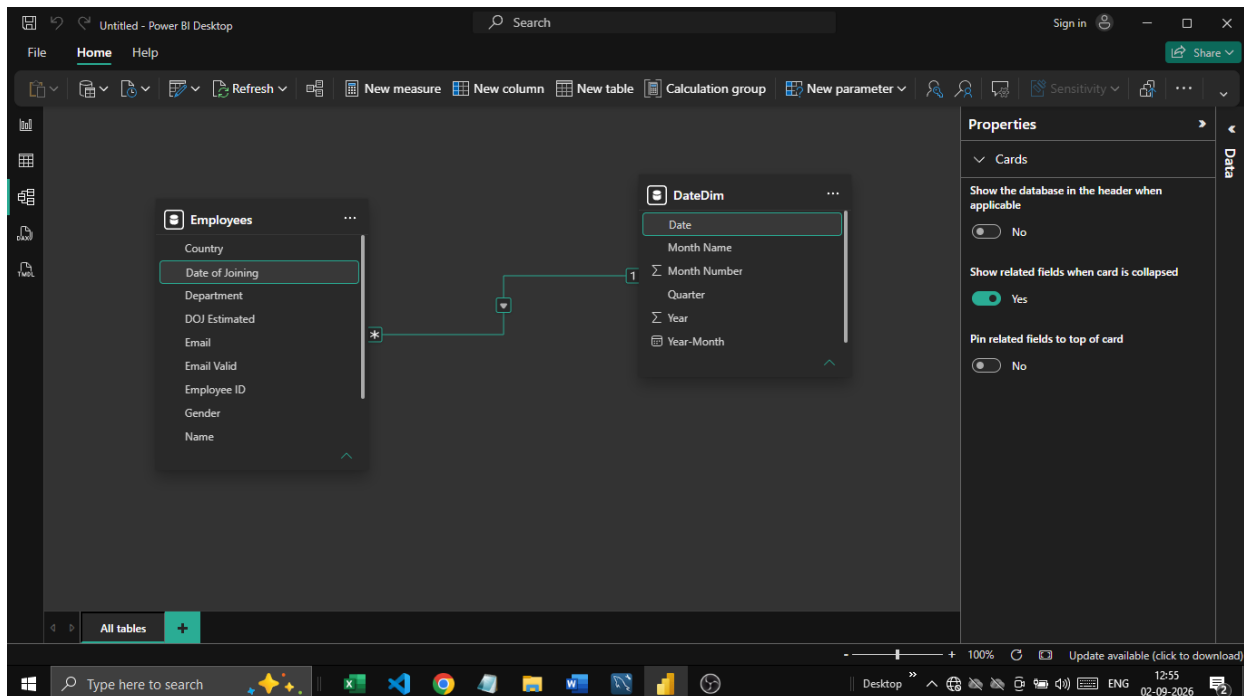
- Trimmed leading/trailing spaces from all text columns.
- Standardized Country and Gender text to consistent Title Case (e.g. "GERMANY" to "Germany").
- Replaced blank/whitespace-only Country values and missing Department/Name/Gender values with "Unknown" so records are not dropped from grouping visuals.
- Converted Salary to a proper numeric type; 8 missing salaries were filled with the median salary of the employee's own Department (a more accurate estimate than an overall average).
- Filled 25 missing Performance Rating values with the overall median rating.

- Date of Joining was converted to a proper Date type. 9 missing dates were flagged with a new "DOJ Estimated" column (True/False) and filled with the dataset's median joining date, so time-based visuals still work correctly while keeping the estimate transparent.
- Validated the Email column with a pattern check; 5 rows contained the placeholder text "invalid_email" - these were replaced with a generated placeholder email (employeeid@company.com) and flagged in a new "Email Valid" column.
- Checked for duplicate rows by Employee ID - none were found.

Data Model

The model uses a star-schema style structure:

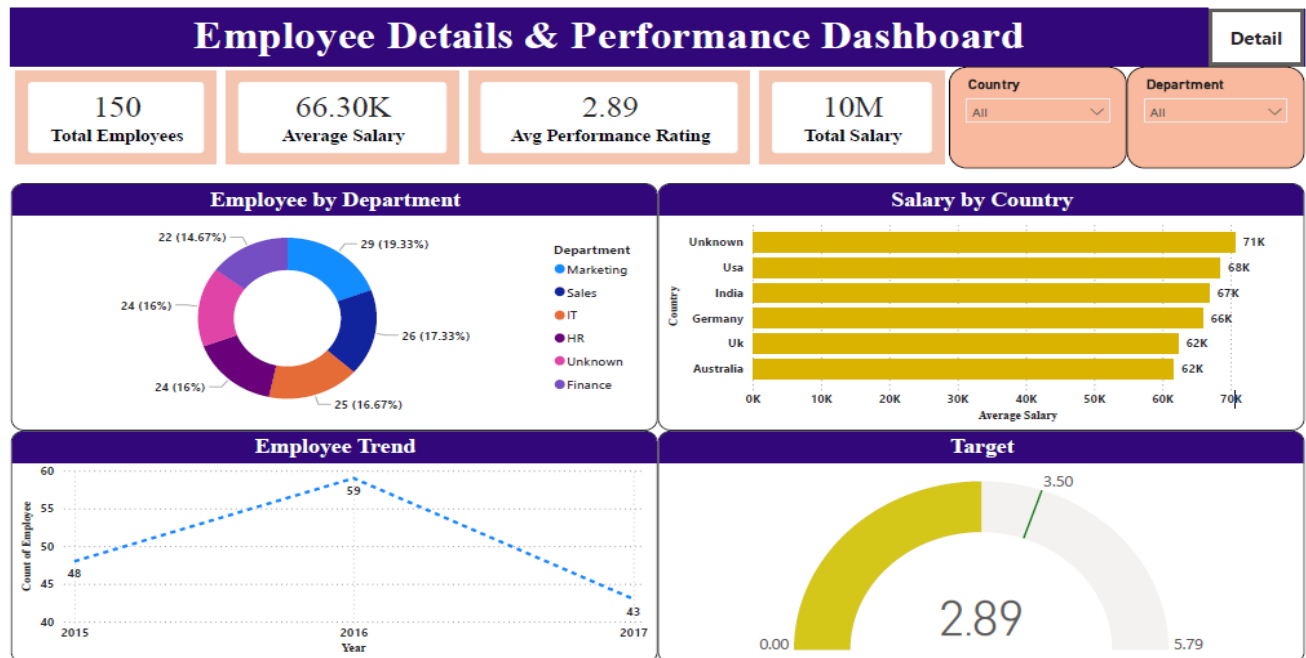
- Employees (Fact/base table) - one row per employee, containing Salary, Performance Rating and Date of Joining.
- DateDim (Dimension table) - a continuous calendar table (Date, Year, Quarter, Month Number, Month Name, Year-Month) built to cover the full Date of Joining range, enabling proper time-intelligence DAX and a Year -> Quarter -> Month drill-down hierarchy.
- Relationship: DateDim[Date] (1) -> Employees[Date of Joining] (many), single direction cross-filter, from DateDim to Employees.



DAX Measures Created

Measure	DAX Formula	Purpose
Total Employees	Total Employees = COUNTROWS(Employees)	Headcount
Total Salary	Total Salary = SUM(Employees[Salary])	Overall payroll cost
Average Salary	Average Salary = AVERAGE(Employees[Salary])	Typical pay level
Min / Max Salary	Max Salary = MAX(Employees[Salary]) Min Salary = MIN(Employees[Salary])	Salary range
Avg Performance Rating	Avg Performance Rating = AVERAGE(Employees[Performance Rating])	Team performance level
Hires This Year	Hires This Year = CALCULATE(COUNTROWS(Employees), YEAR(Employees[Date of Joining]) = YEAR(TODAY()))	Time intelligence - current year hiring
Hires PY	Hires Previous Year = CALCULATE([Hires This Year], SAMEPERIODLASTYEAR(DateDim[Date]))	Comparison across periods
YoY Hiring Growth %	YoY Hiring Growth % = DIVIDE([Hires This Year]-[Hires PY], [Hires PY], 0)	Growth / % change
Salary Rank	Salary Rank = RANKX(ALL(Employees), [Total Salary],, DESC)	Ranking employees by pay
Top 5 Salary Flag	Is Top 5 Salary = IF([Salary Rank] <= 5, "Yes", "No")	Top N analysis
Dept Salary % of Total	Dept % of Total Salary = DIVIDE([Total Salary], CALCULATE([Total Salary], ALL(Employees[Department])))	Department contribution share

Dashboard



Insights Discovered

- Sales has the highest average salary (~₹67,858) followed closely by HR (~₹67,708); Marketing has the lowest average salary (~₹63,789) but the highest average Performance Rating (3.34).
- Employees based in the USA earn the highest average salary (~₹68,426); Australia the lowest (~₹61,610) among rows with a known country.
- Hiring was heaviest in 2016 (59 new joiners) compared to 2015 (48) and 2017 (43), indicating a hiring peak in the middle year of the dataset's date range.
- Performance ratings skew toward the middle: rating 3 is the most common (55 employees), while very few employees are rated 4 (17 employees).
- The top 5 highest-paid employees are spread across Sales, HR, Marketing and IT - high pay is not concentrated in a single department.

Thank You!